# Secure and Scalable LAN Design for UrbanBox Fitness Studio

Student Name: Kamil Salih

Student number: C00307549

Group: Cyber Group B

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Subject: Networking: Switching and VLAN Concepts

# Introduction

UrbanBox is a busy city-centre boxing and fitness studio, popular with local residents and office workers. UrbanBox is opening a new flagship studio and requires a high-performing, secure and reliable wired network for its day-today operations.

I have been asked to design and prototype a local area network (LAN) that will support all essential business functions, ensuring efficient administration, secure data handling and consistent member experiences.

# Departmental Segmentation Using VLANs

The LAN supports multiple departments, each segmented into its own VLAN for traffic isolation, security and management efficiency. The VLANs are set up to optimise network performance while keeping communication secure.

**VLAN Table:**

| Department/Group | VLAN ID | Subnet |
|---|---|---|
| Reception | 10 | 192.168.10.0/24 |
| Finance | 20 | 192.168.20.0/24 |
| Executive | 30 | 192.168.30.0/24 |
| Instructors | 40 | 192.168.40.0/24 |
| Members | 50 | 192.168.50.0/24 |
| Native | 100 | |

Each Switch-to-PC connection has its switchport configured as an access port bound to the corresponding VLAN. The following commands have been used to do this:

switchport mode access

switchport access vlan (vlan ID here)

Each Switch-to-Switch connection has its switchport configured as a trunk port. All of these trunk ports have allowed VLAN lists to prevent any unspecified VLANs from travelling across them. VLAN 100 was set as the native VLAN instead of VLAN 1, in order to reduce the chances of VLAN-hopping. Dynamic Trunking Protocol (DTP) has also been disabled on all trunk ports. The following commands have been used to do this:

switchport mode trunk

switchport trunk allowed vlan 10,20,30,40,50,100

switchport trunk native vlan 100

switchport nonegotiate

Screenshot of VLANs and access port assignments on a switch on the LAN:

**S3#show vlan brief** command output:

```
S3#show vlan brief

VLAN Name                              Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                           active    Po1, Fa0/4, Fa0/5, Fa0/6
                                                 Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                 Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                 Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                 Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                 Fa0/23, Fa0/24, Gig0/1, Gig0/2
10   Reception                         active    Fa0/1, Fa0/2
20   Finance                           active
30   Executive                         active
40   Instructors                       active
50   Members                           active
100  Native                            active
1002 fddi-default                      active
1003 token-ring-default                active
1004 fddinet-default                   active
1005 trnet-default                     active
S3#
```

Screenshot of a trunk configuration on a switch on the LAN:

**S3#show int trunk** command output:

```
S3#show int trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/3       on           802.1q         trunking      100

Port        Vlans allowed on trunk
Fa0/3       10,20,30,40,50,100

Port        Vlans allowed and active in management domain
Fa0/3       10,20,30,40,50,100

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/3       10,20,30,40,50,100
```

# Inter-VLAN Communication and External Connectivity

The chosen method for this network to allow inter-VLAN communication is Router-On-A-Stick.

**Justification for my choice:**

1. Simplifies inter-VLAN routing: only one router interface is in charge of all inter-vlan communication (JumpCloud Inc., 2025).

2. Saves costs: Considering that the flagship studio is new, money must be allocated properly. Therefore, cost-saving methods are recommended in the short term. This one router is aimed towards making the most out of its hardware, so you do not have to get more devices (more routers or layer 3 switches) (JumpCloud Inc., 2025).

3. Easy to configure: considering that all you need to do is configure subinterfaces for one interface, there is not much work to do so this saves time for network administrators at UrbanBox (JumpCloud Inc., 2025).

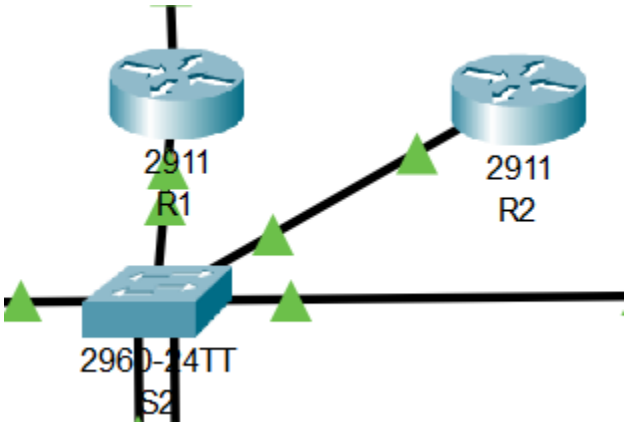
Each subinterface is configured with:

encapsulation dot1Q (vlan ID here)

ip address 192.168.10.254 (HSRP virtual IP)

It is worth noting that I added a backup router that maintains connectivity in case the primary router fails.

Screenshot of main router (R1) and backup router (R2) connected to the central switch (S2):




Screenshot of subinterfaces for the router-on-a-stick:

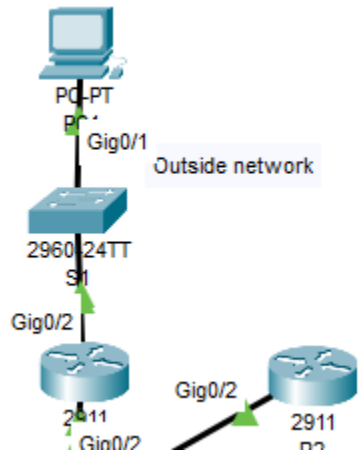
**R1#show ip int brief** command output:

```
R1#show ip int brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.1   unassigned      YES unset  up                    up
GigabitEthernet0/1     unassigned      YES unset  up                    up
GigabitEthernet0/1.1   unassigned      YES unset  administratively down down
GigabitEthernet0/1.2   unassigned      YES unset  administratively down down
GigabitEthernet0/1.10  192.168.10.1    YES manual up                    up
GigabitEthernet0/1.20  192.168.20.1    YES manual up                    up
GigabitEthernet0/1.30  192.168.30.1    YES manual up                    up
GigabitEthernet0/1.40  192.168.40.1    YES manual up                    up
GigabitEthernet0/1.50  192.168.50.1    YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R1#
```

All users should have access to devices and services outside of the gym's network. To demonstrate this, I added a switch and PC outside of the network. The PC is connected to the switch and the switch is connected to the router-on-a-stick.

The external PC has been assigned with an IP within a public IP range (172.32.10.1) (Freda, 2021).

Screenshot of external network:



# IPv4 Addressing with Dynamic Allocation

Each VLAN is assigned a unique /24 IPv4 address range. The network address I chose clearly relates to the VLAN number (for example, VLAN 10 is using 192.168.10.0/24).

IP addresses are allocated dynamically through the router for efficient network management and ease of use. All DHCP pools align with my chose IP IPv4 address ranges.

Each pool includes:

IP address range

Subnet mask
Default Gateway (HSRP virtual IP address)

DNS server (Google's DNS) (Schinkel, no date)

Example of a DHCP pool configuration:

ip dhcp pool VLAN10

network 192.168.10.0 255.255.255.0

default-router 192.168.10.1

dns-server 8.8.8.8

Screenshot of DHCP pool configuration:

**R1#show running-config | section dhcp** command output:

```
R1#show running-config | section dhcp
ip dhcp excluded-address 192.168.10.1 192.168.10.10
ip dhcp excluded-address 192.168.20.1 192.168.20.10
ip dhcp excluded-address 192.168.30.1 192.168.30.10
ip dhcp excluded-address 192.168.40.1 192.168.40.10
ip dhcp excluded-address 192.168.50.1 192.168.50.10
ip dhcp pool vlan10
 network 192.168.10.0 255.255.255.0
 default-router 192.168.10.1
 dns-server 8.8.8.8
ip dhcp pool vlan20
 network 192.168.20.0 255.255.255.0
 default-router 192.168.20.1
 dns-server 8.8.8.8
ip dhcp pool vlan30
 network 192.168.30.0 255.255.255.0
 default-router 192.168.30.1
 dns-server 8.8.8.8
ip dhcp pool vlan40
 network 192.168.40.0 255.255.255.0
 default-router 192.168.40.1
 dns-server 8.8.8.8
ip dhcp pool vlan50
 network 192.168.50.0 255.255.255.0
 default-router 192.168.50.1
 dns-server 8.8.8.8
R1#
```

Screenshot of Client IP Assignment verification:

PC6's **ipconfig /all** command output:

```
C:\>ipconfig /all

FastEthernet0 Connection:(default port)

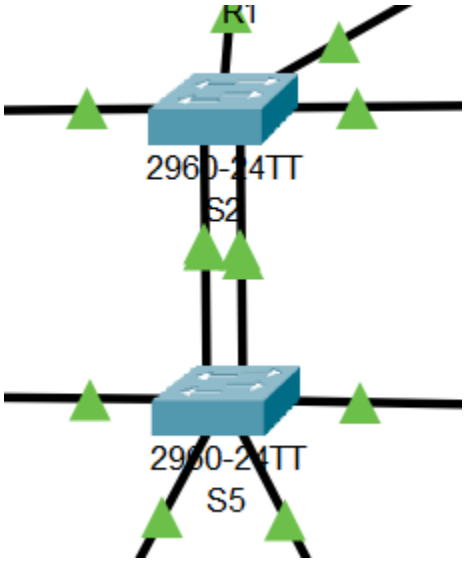
   Connection-specific DNS Suffix..:
   Physical Address.................: 0090.0CC4.5C1B
   Link-local IPv6 Address.........: FE80::290:CFF:FEC4:5C1B
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.30.11
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     192.168.30.1
   DHCP Servers....................: 192.168.30.1
   DHCPv6 IAID.....................:
   DHCPv6 Client DUID..............: 00-01-00-01-57-7D-55-AA-00-90-0C-C4-5C-1B
   DNS Servers.....................: ::
                                     8.8.8.8
```

# Network Redundancy for High Availability

## Layer 2 Redundancy

EtherChannel has been established between switch S2 and switch S5. It turns two links and turns them into one logical interface, which increases bandwidth as well as providing failover.

Screenshot of Etherchannel:

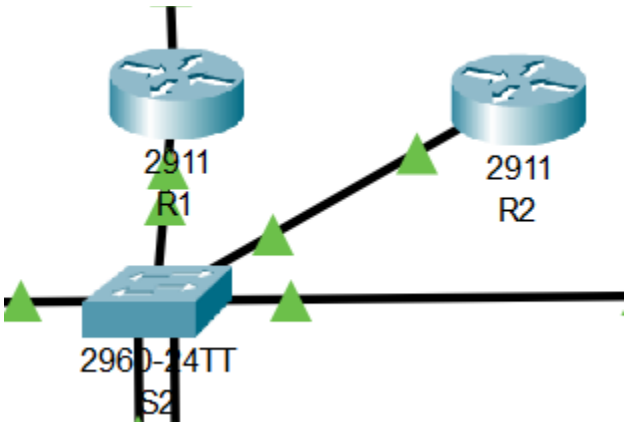


## Layer 3 Redundancy

HSRP has been configured on both routers. The router-on-a-stick is active, while the other is on standby.

If the active router fails, the standby router takes over. In other words, inter-vlan communication will not be interrupted. The devices still have the same default gateway due to the two routers sharing the same virtual IP address.

Screenshot of the two routers connected to the central switch:



# Network Device Security and Hardening

Security measures implemented:

**Passwords:** The password ‘cisco’ has been assigned to the physical console ports of the routers and switches (password cisco and login). The password ‘class’ has been assigned for privileged EXEC mode (enable secret class). Obviously in a real-world scenario, different passwords would be assigned but for the sake of simplicity for the project, these passwords have been assigned.

**Disabled DTP:** Automatic negotiation of trunking links between two Cisco switches is not possible anymore on any switch. This prevents security vulnerabilities such as VLAN hopping.

**Native VLAN is set to a non-default value:** Instead of the native VLAN being VLAN 1, it is now VLAN 100.

## Screenshots of security configuration:

## **S2# show running-config** command output:

```
!
!
!
line con 0
 password class
 login
!
```

```
interface GigabitEthernet0/1
 switchport trunk native vlan 100
 switchport trunk allowed vlan 10,20,30,40,50,100
 switchport mode trunk
 switchport nonegotiate
!
interface GigabitEthernet0/2
 switchport trunk native vlan 100
 switchport trunk allowed vlan 10,20,30,40,50,100
 switchport mode trunk
 switchport nonegotiate
!
```

## **S2# show int fa0/4 switchport** command output:

```
S2#show int fa0/4 switchport
Name: Fa0/4
Switchport: Enabled
Administrative Mode: trunk
Operational Mode: trunk
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: dot1q
Negotiation of Trunking: Off
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 100 (Native)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: 10,20,30,40,50,100
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
```

# Commands for Verification

**VLANs:** show vlan brief

**Trunks:** show interfaces trunk

**Router-on-a-Stick Subinterfaces:** show ip interface brief

**DHCP:** show ip dhcp binding

**DHCP on PC:** ipconfig /all

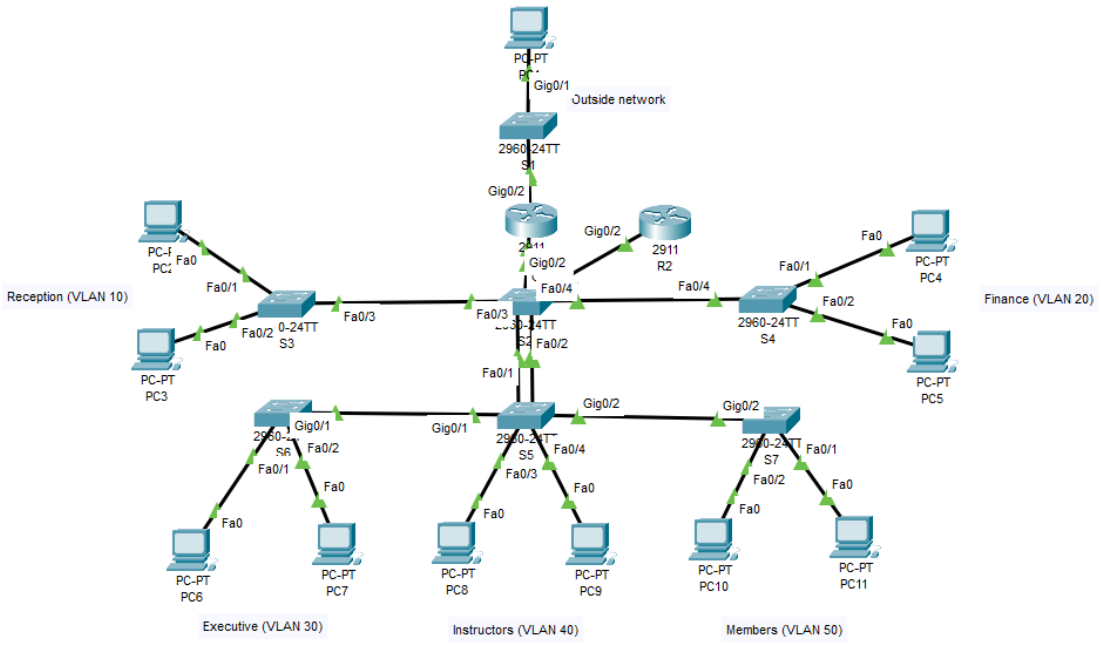
**Inter-VLAN Routing:** show ip route

**STP Redundancy:** show spanning-tree

**Device Health:** show running-config

**Layer 2 switching configuration:** show interfaces (interface_id) switchport

# Screenshot of the entire topology



# References

JumpCloud Inc. (2025) *What Is a Router-on-a-Stick?.* Available at: **https://jumpcloud.com/it-index/what-is-a-router-on-a-stick** (Accessed 22nd November 2025)

Freda, A. (2021) *What's the difference between public and private IP addresses?.* Available at: **https://www.avg.com/en/signal/public-vs-private-ip-address** (Accessed 22nd November 2025)

Schinkel, L. (no date) *What Is Google DNS? Changing Your DNS to 8.8.8.8.* Available at: https://www.whatismyip.com/google-dns/ (Accessed 22nd November 2025)